Unit 5: Photo Chemistry

Part 1 - Introduction

Jabłoński diagram

Aleksander Jabłoński was a Polish physicist who specialized in molecular spectroscopy and photophysics. He proposed the Jabłoński diagram in the 1930s to explain how molecules behave after absorbing light.

- Before Jabłoński's work, processes like fluorescence and phosphorescence were difficult to explain clearly.
- The Jabłoński diagram is now fundamental in photochemistry, photophysics, spectroscopy, fluorescence microscopy, and laser science.
- Jabłoński conducted much of his scientific work during World War II and the post-war period in Poland, when research resources were very limited.

Sometimes the most powerful scientific ideas are simple ways of explaining complex phenomena. - A Single Idea Can Influence Generations

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Introduction

How does a chemical reaction occurs?

- A chemical reaction occurs when moving molecules collide, break their bonds, and exchange atoms to form new products.
- In a chemical reaction, activation energy is the minimum amount of energy required for the reactants to be converted into products.

$$A \xrightarrow{\Delta} B$$

Thermal reaction

Activation energy obtained by random intermolecular collision on the increase of tempertaure

$$C \xrightarrow{h\nu} D$$

Photochemical reaction

Activation energy is obtained by absorption of photons of light of a particular energy

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Photophysical and Photochemical Processes

Photochemistry is defined as that branch of chemistry which deals with the process involving emission or absorption of radiation.

1. Photophysical Processes

- Processes take place in the presence of light but do not involve any chemical reaction. They take place due to absorption of light by the substances followed by the emission of the absorbed light.
- Photophysical processes include fluorescence, phosphorescence and photoelectric effect.

2. Photochemical Reactions

- These are the reactions that take place by absorption of radiations of suitable wavelength.
- In such processes, the absorbed light energy is first stored in the molecule and then it is further used to bring about the reaction.

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Difference between Photochemical and Thermochemical Reactions

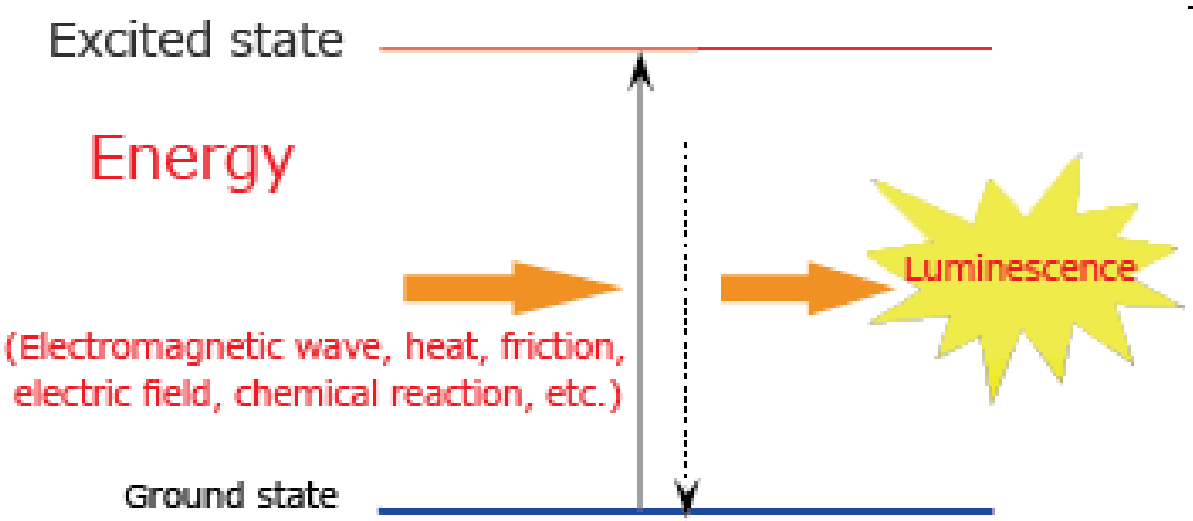
Photochemical Reactions	Thermochemical Reactions
1. These involve absorption of light radiations.	1. These reactions involve absorption or evolution of heat.
2. The presence of light is the primary requirement for reactions to take place.	2. These reactions can take place in dark as well as in light.
3. Temperature has a very little effect on the rate of photochemical reactions.	3. Temperature has a significant effect on the rate of a thermochemical reaction.
4. ΔG for photochemical spontaneous reactions may be +ve or –ve.	4. ΔG for a thermochemical reaction is always negative.
5. Photochemical activation is highly selective. The absorbed photon excites a particular atom or group of atoms which become site for the reaction.	5. Thermochemical activation is not selective in nature.

Photophysical processes

Luminescence

The glow produced by a body, without the involvement of heat is called luminescence. It is also called cold light.

We know when a block of iron is heated, it first becomes red hot and eventually becomes white and begins to glow. But this will not be called luminescence. Luminescence does not occur by the action of heat. It can occur, for example, when the electrons excited to higher levels by the absorption of radiation return to original levels.



Types of luminescence

- Chemiluminescence
- Photoluminescence
- Bioluminescence
- Electroluminescence
- Thermoluminescence
- Cathodoluminescence, etc.

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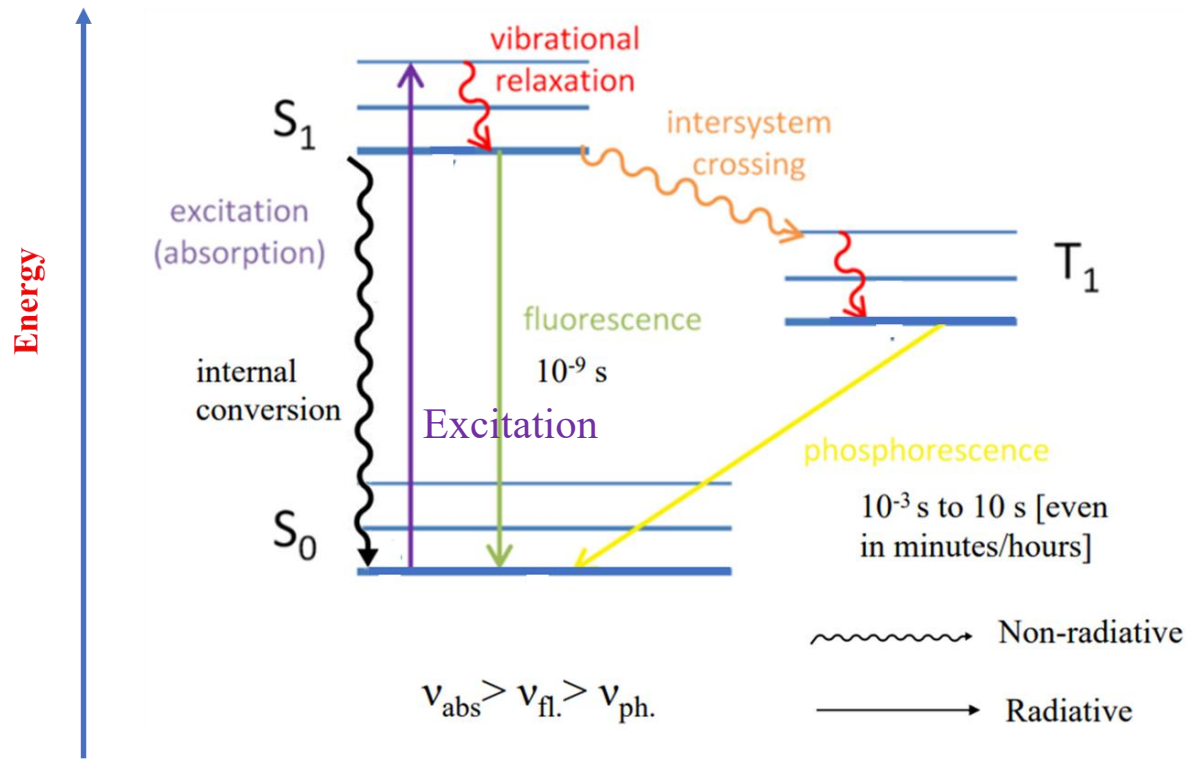
Jablonski Diagram (possible consequences of applying photons from the visible spectrum of light to a particular molecule)

- **Absorbance** is the method by which an electron is excited from a lower energy level to a higher energy level.

- **Vibrational relaxation** is a non-radiative process. The energy deposited by the photon into the electron is given away to other vibrational modes as kinetic energy



- **Internal conversion:** The excited electron is transferred from a vibration level in one electronic state to another vibration level in a lower electronic state

- **Fluorescence (allowed):** The molecule emits a photon and goes to the ground state. The energy of fluorescent photons is always less than that of the exciting photons. This difference is because energy is lost in internal conversion and vibrational relaxation

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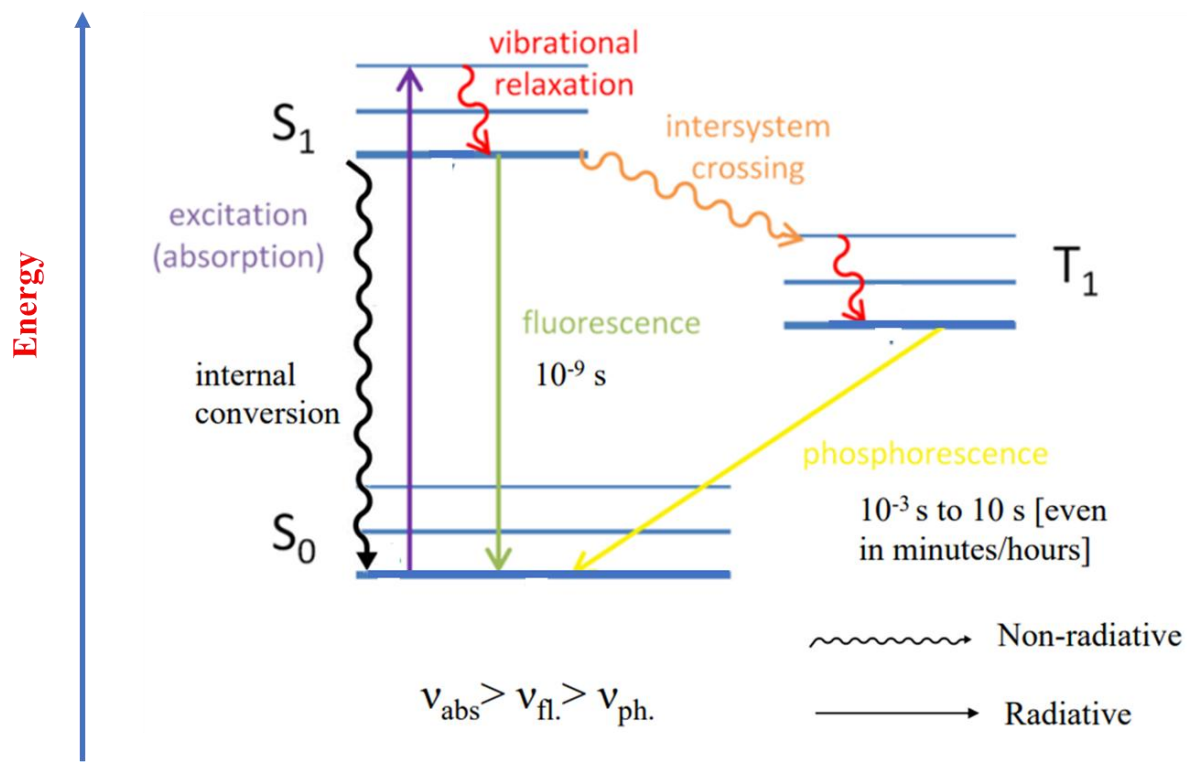
Jablonski Diagram (possible consequences of applying photons from the visible spectrum of light to a particular molecule)

Intersystem Crossing: The electron changes spin multiplicity from an excited singlet state to an excited triplet state. (forbidden, but allowed due to vibronic coupling)

Phosphorescence: A radiative transition from an excited triplet state to a singlet ground state (very slow, forbidden transition)
Delayed fluorescence: The transition back to the first excited singlet level, leading to the emitting transition to the ground state.



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Chemiluminescence

When a photochemical reaction takes place, light is absorbed. However, there are certain reactions in which light is produced. The emission of light in chemical reactions at ordinary temperatures is called chemiluminescence. The chemiluminescence is just the reverse of a photochemical reaction.

Reason: In a chemiluminescent reaction, the energy released in the reaction makes the product molecule electronically excited. The excited molecule then gives up its excess energy as visible light while reverting to ground state

Examples:

- The oxidation of yellow phosphorus in oxygen or air to give P_2O_5 is accompanied by the emission of visible greenish-white luminescence.

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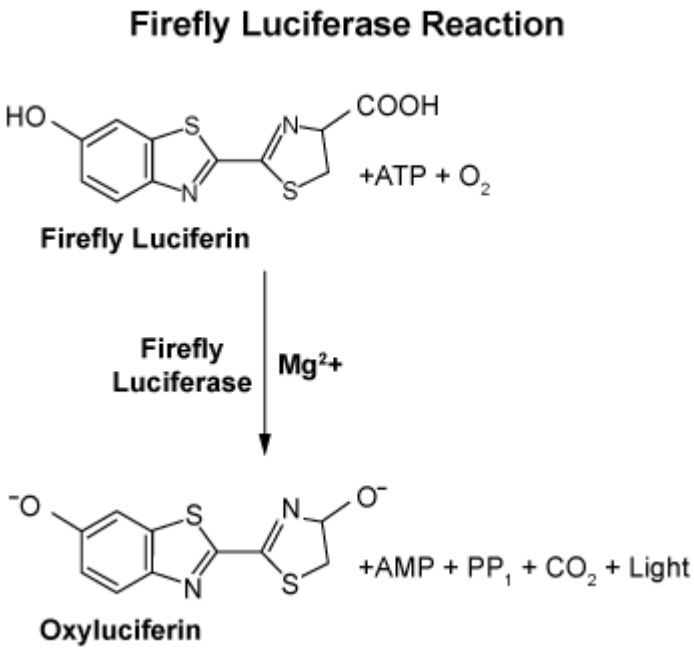
Bioluminescence

When chemiluminescence occurs in a living organism, it is called bioluminescence. The best-characterized bioluminescence system is that of the fireflies.

The light emitted by glow worms is due to the oxidation of the protein, luciferin, apresent in glow worms.



- Firefly luciferase catalyzes a two step reaction
- First, firefly luciferin reacts with adenosine-5′-triphosphate (ATP) leading to the formation of an adenylyl intermediate
- Subsequently, this intermediate will be oxidized in the presence of molecular oxygen to form excited state oxyluciferin, which is the light emitter.

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Fluorescence

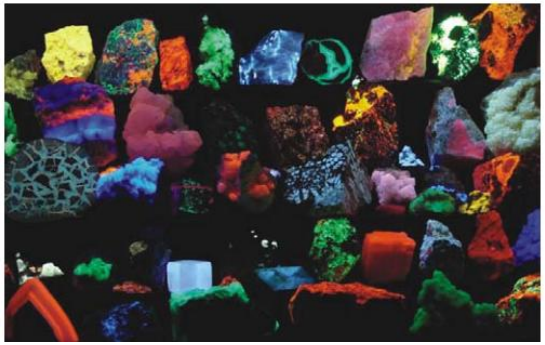
There are certain substances which when exposed to light of certain radiations absorb the energy and then immediately or instantaneously start re-emitting the energy. Such substances are called fluorescent substances and the phenomenon is called fluorescence. Fluorescence starts as soon as the substance is exposed to light and it stops as soon as the light is cut off

Examples:

(a) Fluorite, CaF_2

(b) Certain organic dyes such as eosin, fluorescein, etc.

(c) Certain inorganic compounds such as uranyl sulphate, UO_2SO_4



Fluorescent minerals, shown under ultraviolet light

Reason: The absorption of energy results into the excitation of the electrons followed immediately by jumping back of the excitated electrons to the lower levels. As a result, the absorbed energy is emitted back.

Wavelength: The wavelength of the emitted light is usually greater than that of die absorbed light.

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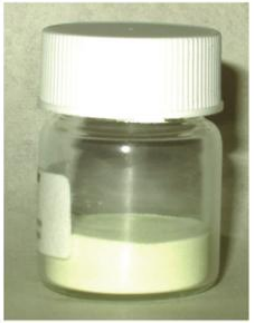
Phosphorescence

There are certain substances which continue to glow for some time even after the external light is cut off. Such substances are called phosphors or phosphorescent substances and the phenomenon is called phosphorescence. Thus, in a way, phosphorescence is slow fluorescence.

Examples.
(a) zinc sulphide and sulphides of the alkaline earth metals.



Phosphorescence is slow fluorescence. It is found mostly in solids, because the molecules have least freedom of motion. The excited electrons thus keep on jumping back slowly for quite some time.

It has been found that fluorescent substances become phosphorescent if fixed by suitable methods, e.g. by fusion with other substances, etc. For example, many dyes show fluorescence when dissolved in fused boric acid or glycerol and when cooled to rigid mass become phosphorescent.

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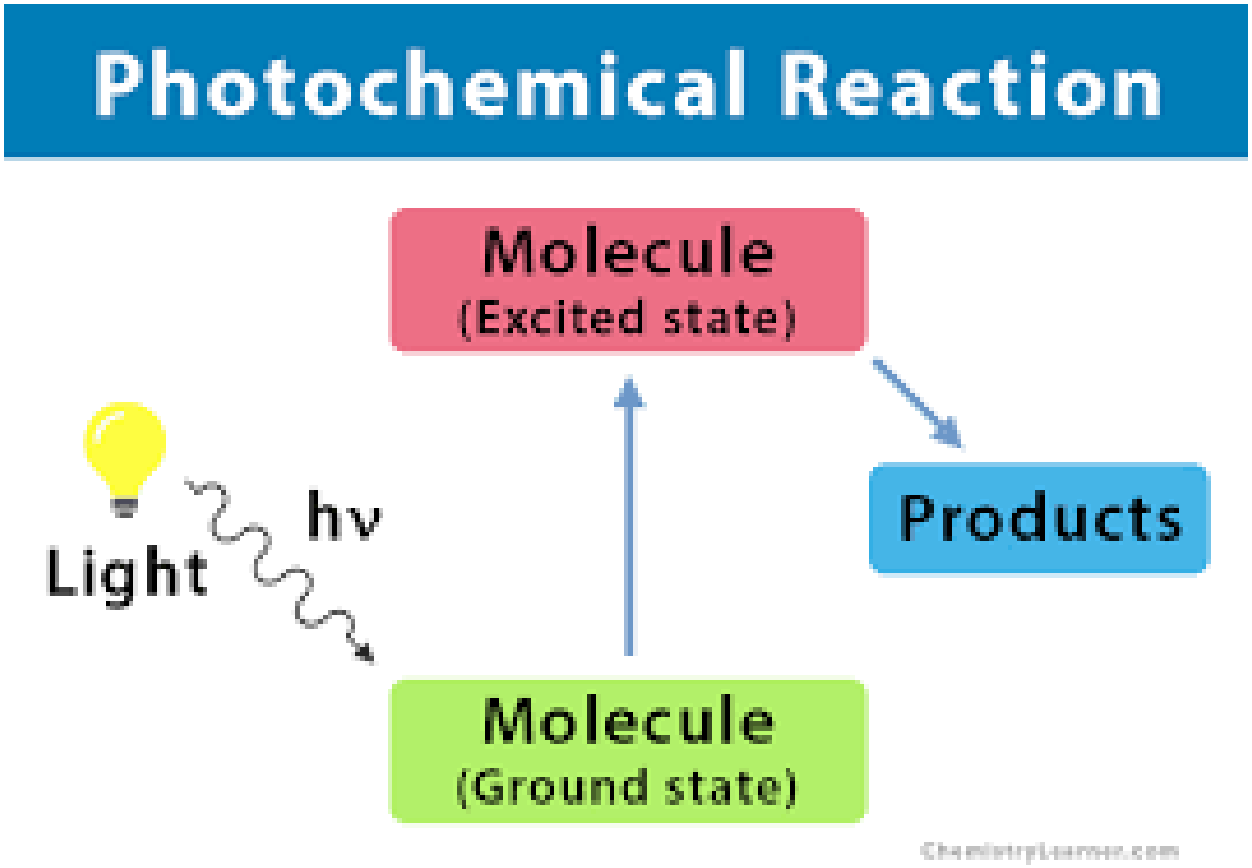

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Demonstration of a Photochemical reaction

A mixture of hydrogen and chlorine remains unchanged with lapse of time. But when exposed to light, the reaction occurs with a loud explosion

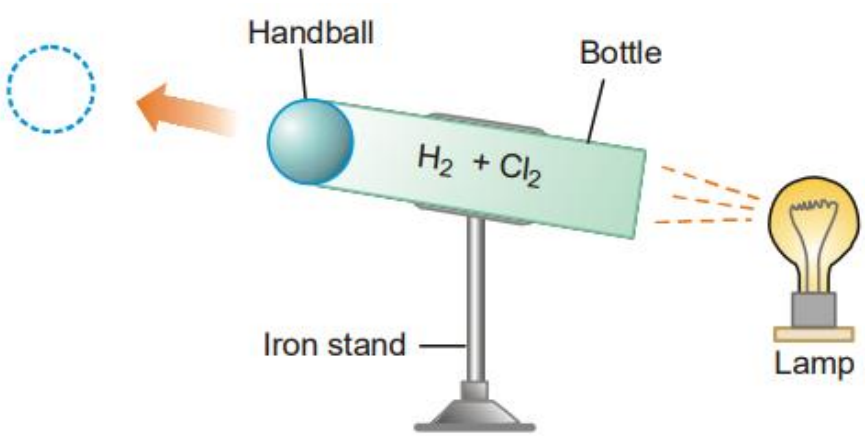
$$H_2 + Cl_2 \xrightarrow{\text{dark}} \text{No reaction}$$

$$H_2 + Cl_2 \xrightarrow{\text{light}} 2\,HCl$$

The 'HCl-cannon' experiment.

- A bottle is filled with equimolar amounts of hydrogen and chlorine and tightly stoppered with a handball.

- When the lamp is turned on, a beam of light falls on the mixture through the bottom of the bottle and the reaction occurs with an explosion.



- The ball is expelled with high velocity so that it strikes the opposite wall

Unit 5: Photochemistry

Laws of Photochemistry

1. Grotthus Law (First Law of Photochemistry)

When light falls on a body, a part of it is reflected, a part of it is transmitted and the rest of it is absorbed. It is only the absorbed light which is effective in bringing about chemical reaction

- It does not imply that the absorbed light must always result into chemical reaction. The absorbed light may simply bring about phenomena such as fluorescence, phosphorescence, etc.

- The absorbed light energy may be simply converted into thermal energy, e.g. in case of potassium permanganate solution, the light energy is absorbed strongly but no chemical effect is produced.

- In some cases, it is observed that light energy may not be absorbed by the reacting substance directly but may be absorbed by some other substance present along with the reacting substance. The energy thus absorbed is then passed onto the reacting substance which then starts reacting. This process is called ***photosensitisation***. Photosynthesis of carbohydrates in plants where chlorophyll acts as a photosensitiser is a prominent example of photosensitisation.

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2. Stark-Einstein Law of Photochemical Equivalence (Second Law of Photochemistry)

- According to this law, for each photon of light absorbed, only one molecule is activated for the subsequent reaction.
- This photo equivalence law was derived by Einstein during his development of the quantum theory of light; hence it is known as the Stark-Einstein law.

Every atom or molecule that takes part in a photochemical reaction absorbs one quantum of the radiation to which the substance is exposed

The energy possessed by one mole of photons or the energy absorbed by one mole of the reacting molecules is called one Einstein

Numerical Value of Einstein $E = Nh\nu$ N is the Avogadro's number

$$E = \frac{Nhc}{\lambda} \quad \text{since} \quad \nu = \frac{c}{\lambda}$$

$$E = \frac{2.859}{\lambda} \times 10^5 \ \text{kcal mol}^{-1}$$

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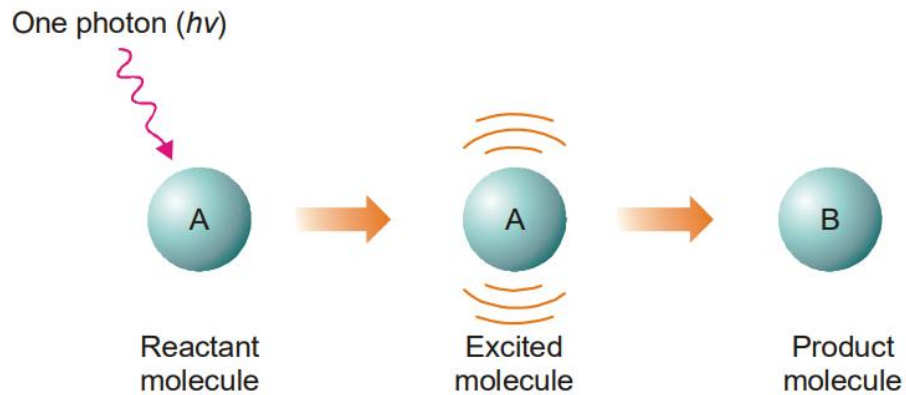
2. Stark-Einstein Law of Photochemical Equivalence (Second Law of Photochemistry)

- A molecule 'A' absorbs a photon of radiation and gets activated.
- The activated molecule (A*) then decomposes to yield B.



$$A + h\nu \longrightarrow A^*$$

$$A^* \longrightarrow B$$

$$\text{overall} \quad A + h\nu \longrightarrow B$$

In practice, we use molar quantities. That is, one mole of A absorbs one mole of photons or one einstein of energy, E.

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3. Beer-Lambert Law

The Beer-Lambert law proposed that the absorption of light or single wave length is related to the thickness of the absorbing compound and the optical path of light

$$I_t = I_o e^{-an}$$

where I_o = intensity of incident light
I_t = intensity of transmitted light
a = constant
n = thickness of absorbing material.

The above equation is subsequently modified as follows,

$$I = I_o e^{-\epsilon cl}$$

$$\frac{I}{I_o} = e^{-\epsilon cl}$$

$$\log \frac{I_o}{I} = \epsilon cl \quad \text{(or)} \quad A = \epsilon cl$$

A = Optical density (or) Absorbance (how much light is absorbed) ; *l* = path length;
c = concentration of the sample;
ε = molar extinction coefficient or molar absorptivity and is a measure of the probability of the electronic transition

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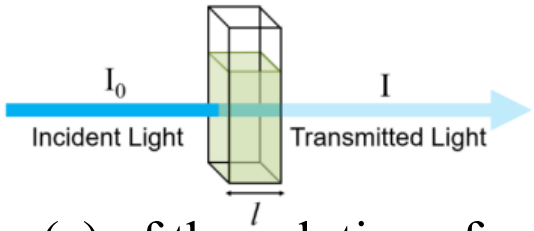
Beer- Lambert Law

The Beer-Lambert law relates the attenuation of light to the properties of the material through which the light is traveling.



1. The absorbance is directly proportional to the concentration (c) of the solution of the sample used in the experiment.
2. The absorbance is directly proportional to the length of the light path (l), which is equal to the width of the cuvette

$$A = log\frac{Io}{I} = \varepsilon cl$$

A = Optical density (or) Absorbance (how much light is absorbed) ; *l* = path length;
c = concentration of the sample;
ε = molar extinction coefficient or molar absorptivity and is a measure of the probability of the electronic transition

The larger the molar absorptivity, the more probable the electronic transition.
In uv spectroscopy, the concentration of the sample solution is measured in the units mol L^{-1} and the length of the light path in cm. Thus (absorbance is unitless) the units of molar absorptivity are L mol^{-1} cm^{-1}. (mostly unitless)

If ε (at λ_{max}) is known, we can measure A (at λ_{max}), path length is typically ~1 cm, then c can be calculated.

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Photosensitized Reactions

A photosensitized reaction is an indirect photochemical reaction where molecules of one chemical species (photosensitizer) absorb light and cause chemical changes in molecules of another chemical species.

Photosensitization

$$D \xrightarrow{hv} D^* \quad \text{(Excited donor molecule)}$$

$$A + D^* \longrightarrow A^* \quad \text{(Excited acceptor molecule)}$$

$$A^* \longrightarrow \text{product/s}$$

Photosensitizer/Donor molecule = (D)

Acceptor molecule = A

Examples of Photosensitized reactions

Reaction between H_2 and O_2 to form H_2O_2

$$Hg + hv \longrightarrow Hg^* \quad \text{Primary absorption}$$

$$Hg + H_2 \longrightarrow 2H + Hg \quad \text{Energy transfer}$$

$$\left.\begin{array}{l} H + O_2 \longrightarrow HO_2 \\ HO_2 + HO_2 \longrightarrow H_2O_2 + O_2 \end{array}\right\} \text{Reaction}$$

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Quantum Efficiency (or) Quantum Yield: ϕ

To relate the number of quanta absorbed with the number of reacting molecules, a term called quantum efficiency or quantum yield (ϕ) has been introduced

The number of molecules reacted or formed per photon of light absorbed is termed Quantum yield. It is denoted by φ

$$\phi = \frac{\text{No. of molecules reacted or formed}}{\text{No. of photons absorbed}}$$

- For a reaction that obeys strictly the Einstein law, one molecule decomposes per photon, the quantum yield $\varphi = 1$.
- When two or more molecules are decomposed per photon, $\varphi > 1$ and the reaction has a high quantum yield.
- If the number of molecules decomposed is less than one per photon, the reaction has a low quantum yield.

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Cause of high quantum yield

When one photon decomposes or forms more than one molecule, the quantum yield φ > 1 and is said to be high

(a) Reactions subsequent to the Primary reaction

$$AB + h\nu \longrightarrow A + B \quad \text{Primary}$$
$$AB + A \longrightarrow A_2 + B \quad \text{Secondary}$$

The excited atoms that result may start a subsequent secondary reaction

(b) A reaction chain forms many molecules per photon

$$A_2 + h\nu \longrightarrow 2A \quad ...(1) \quad \text{Primary}$$
$$A + B_2 \longrightarrow AB + B \quad ...(2)$$
$$B + A_2 \longrightarrow AB + A \quad ...(3)$$

Secondary Reaction chain

The excited atom that is produced starts a secondary reaction chain

Examples of high quantum yield

$$HI + h\nu \longrightarrow H + I \quad ...(1) \quad \text{Primary}$$
$$H + HI \longrightarrow H_2 + I \quad ...(2)$$
$$I + I \longrightarrow I_2 \quad ...(3)$$

Secondary

$$2HI + h\nu \longrightarrow H_2 + I_2 \quad \text{Overall reaction}$$

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Causes of low quantum yield

(a) Deactivation of reacting molecules.

The excited molecules in the primary process may be deactivated before they get opportunity to react. This is caused by collisions with some inert molecules or by fluorescence.

$$A + h\nu \longrightarrow A^* \qquad \textit{Activation}$$

$$A^* \longrightarrow A + h\nu' \qquad \textit{Fluorescence}$$

(b) Occurrence of reverse of primary reaction

The primary reaction generally yields a polymer. The product then undergoes a thermal reaction giving back the reactant molecules

$$2A \underset{\text{thermal}}{\overset{h\nu}{\rightleftharpoons}} A_2$$

The reverse thermal reaction proceeds till the equilibrium state is reached

Examples of low quantum yield

$$2C_{14}H_{10} \underset{\text{thermal}}{\overset{h\nu}{\rightleftharpoons}} C_{28}H_{20}$$

When anthracene, $C_{14}H_{10}$, dissolved in benzene is exposed to ultraviolet light, it is converted to dianthracene, $C_{28}H_{20}$

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Primary and Secondary reactions

The overall photochemical reaction may consist of :

(a) A primary reaction proceeds by absorption of radiation.

(b) A secondary reaction is a thermal reaction which occurs subsequent to the primary reaction.

For example, the decomposition of HBr occurs as follows :

$HBr + h\nu \longrightarrow H + Br$		Primary reaction
$HBr + H \longrightarrow H_2 + Br$		Secondary reaction
$Br + Br \longrightarrow Br_2$		Secondary reaction
$2HBr + h\nu \longrightarrow H_2 + Br_2$		Overall reaction

Evidently, the primary reaction only obeys the law of photochemical equivalence strictly. The secondary reactions have no concern with the law.

THE SUCCESSFUL WARRIOR IS THE AVERAGE MAN, WITH LASER-LIKE FOCUS.

BRUCE LEE